Career Nexus : Resume-Job Matching System

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Introduction

- A job portal that connects job seekers and employers through intelligent resume–job matching .

- Using cosine similarity, it delivers accurate recommendations , smart search and an intuitive interface for candidates ,employers , and administrators alike.

- It is built using the MERN stack : MongoDB, Express.js, React.js, and Node.js, which makes the system fast, easy to use, and able to grow as more users join.

Problem Statement

Drawbacks of Traditional Job Portals

- Fail to match resume and job description accurately
- Lack of personalization
- Employers struggle to filter and identify the most suitable candidates

Objectives

- Develop a full-featured job portal with role-based access and control (for job seekers, employers, and admins) that integrates AI-driven resume–job matching using cosine similarity.

- Provide personalized job recommendations to users by analyzing their profiles, skills, and experience, ensuring more accurate and relevant job suggestions.

Scope

- Users can create profiles, upload resumes, and browse or apply for jobs.
- Employers can post job openings and filter applicants.
- The system matches resumes and job listings using cosine similarity for relevance.
- Personalized job suggestions are shown to job seekers based on skills and experience.
- Admins can monitor activities, manage users, and control job listings.
- Notification system alerts users about new jobs, applications, and updates.

Limitation

- The system requires stable internet connectivity for real-time features.
- Resume matching accuracy depends on resume formatting and keyword clarity.
- It does not support video interviews or built-in chat between users.
- Recommendations may not be 100% accurate for every user.
- Currently, only English-language resumes and listings are supported.

Literature Review

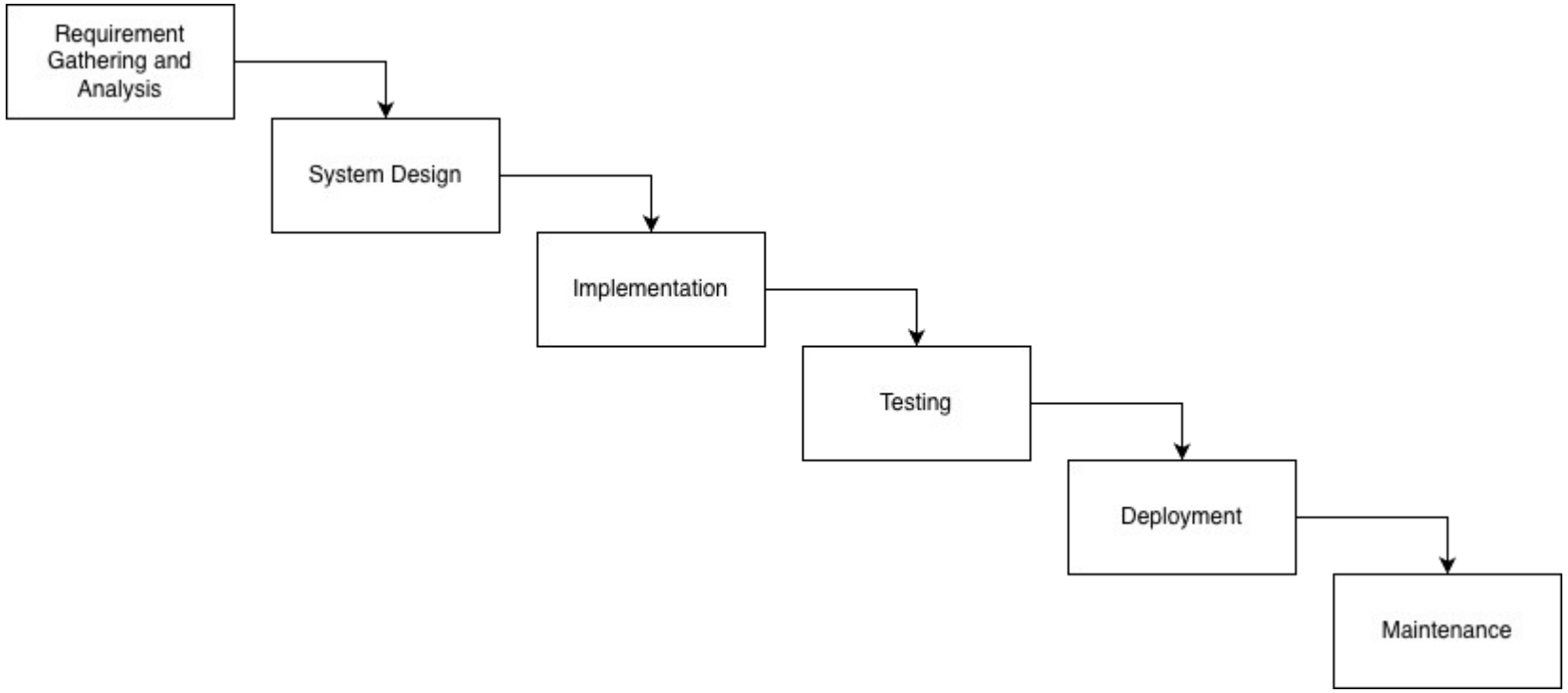
- Cosine similarity has been widely applied in job recommendation systems like Career Nexus to improve resume matching by capturing semantic relationships between job descriptions and candidate profiles. Its ability to measure the cosine of the angle between vectors provides a normalized metric that is particularly effective for high-dimensional text analysis, making it suitable for comparing diverse resume formats.

- Research highlights its superiority over Jaccard similarity and Euclidean distance in recommendation systems, as it focuses on directional alignment rather than set overlap or magnitude differences .

- In Career Nexus, cosine similarity facilitates personalized job recommendations to analyze candidate skills and preferences.

Development Methodology

Waterfall Model



Requirement Analysis

- Functional requirements
- Non-functional requirements

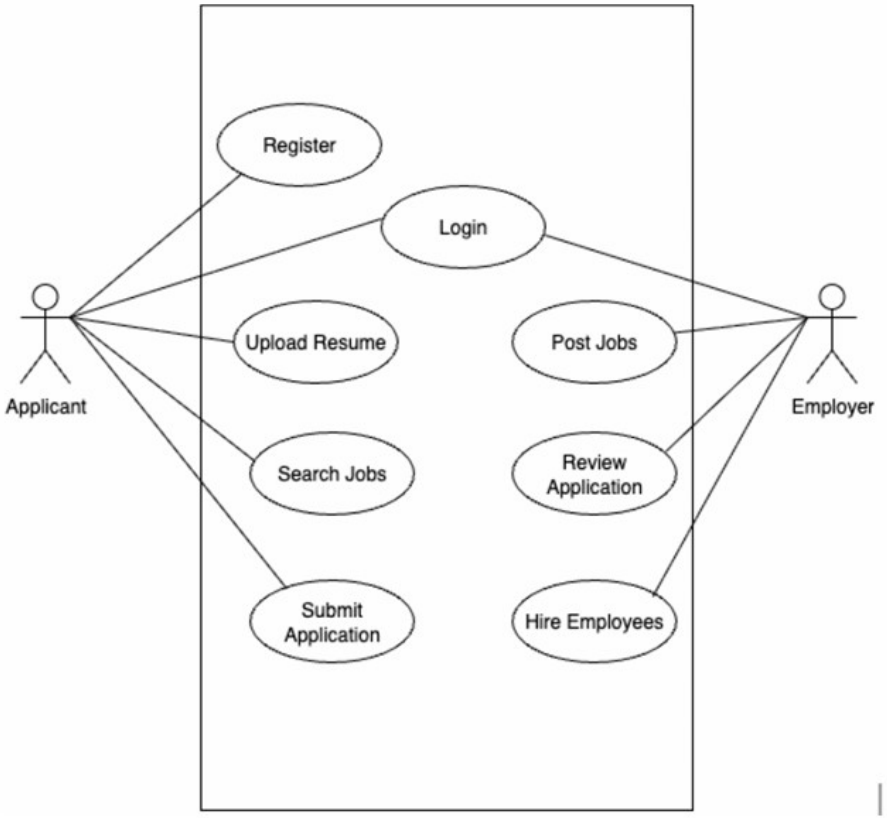
Functional Requirement

- **User Registration and Login:**
  The system shall allow both job seekers and employers to register and log in securely using valid credentials.
- **Job Posting:**
  Employers shall be able to create and publish job listings with details such as title, description, requirements, and deadlines.
- **Resume Upload:**
  Job seekers shall be able to upload resumes in supported formats (PDF or DOCX) for storage and matching.
- **Job Search:**
  The system shall allow users to search for jobs using relevant keywords.
- **AI-Based Resume–Job Matching:**
  The system shall use the cosine similarity algorithm to compare resumes and job descriptions to recommend the most suitable matches.
- **Job Application Submission:**
  Job seekers shall be able to apply for available jobs through the portal.
- **Employer Dashboard:**
  Employers shall have access to a dashboard to view, manage, and shortlist applicants.
- **Notification System:**
  The system shall send notifications (via email or app alerts) to users regarding job applications, interview schedules, and updates.
- **Admin Management:**
  The system shall allow administrators to manage user accounts, approve job postings, and configure overall system settings**.**

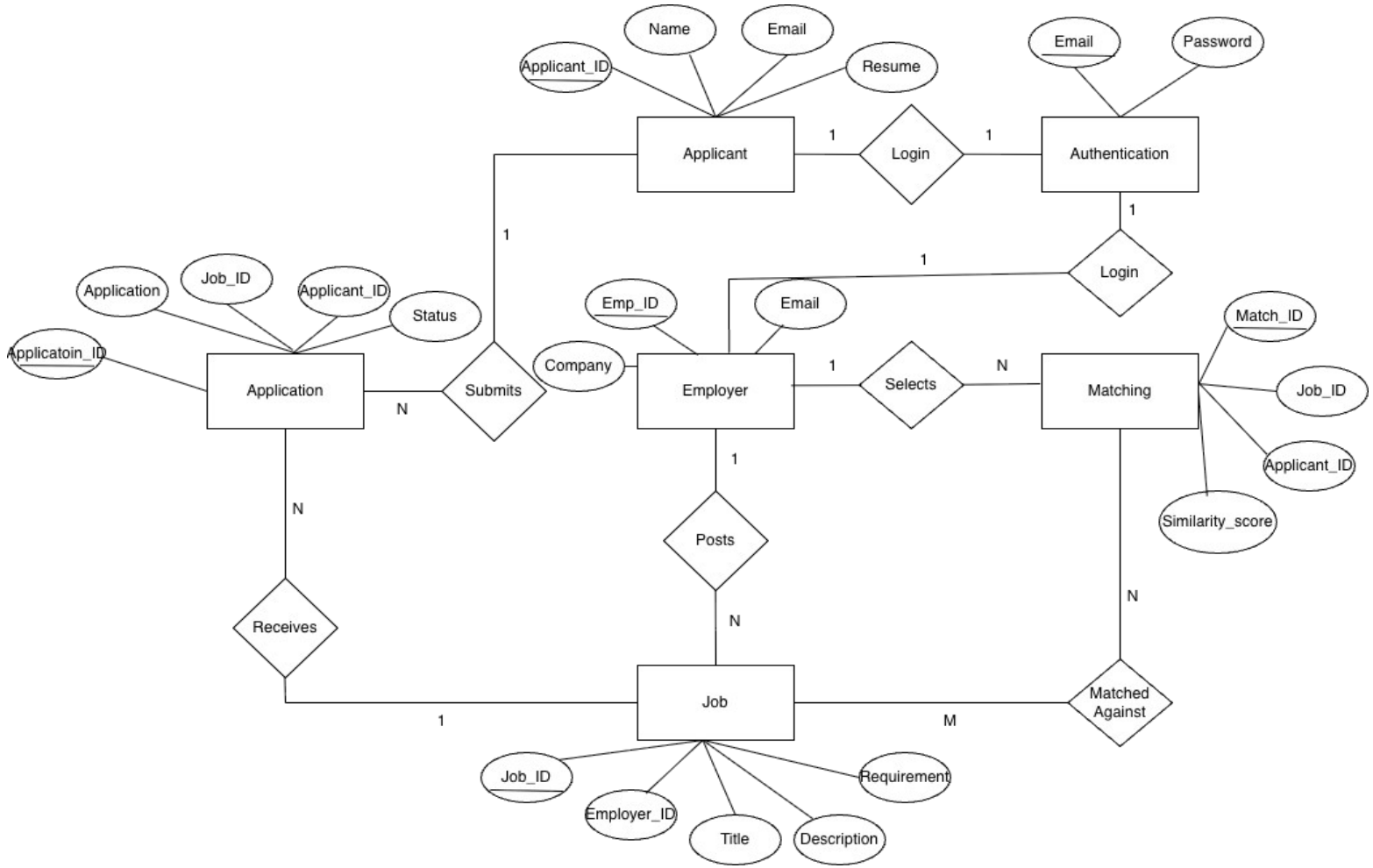
Non-functional Requirement

- **Scalability:**
  The system shall efficiently handle a growing number of users and job postings without any noticeable drop in performance**.**
- **Security:**
  User data, including login credentials and resumes, shall be protected using encryption and must comply with data privacy and security standards.
- **Usability:**
  The user interface shall be simple and intuitive, with a response time of less than 2 seconds, ensuring a smooth and user- friendly experience.
- **Availability:**
  The platform shall be available 24/7, with minimal downtime for maintenance or updates.
- **Performance:**
  The cosine similarity–based matching algorithm shall process resume–job comparisons within a few seconds, even for hundreds of records.
- **Compatibility:**
  The website shall be fully responsive and functional across all major devices and screen sizes, including PCs, tablets, and mobile phones.
- **Maintainability:**
  The system’s codebase shall be well-documented and structured, enabling future developers to easily understand, update, and fix issues

Use Case Diagram

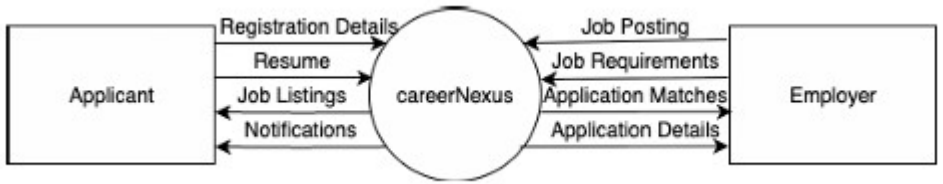


Data Modeling: Entity-Relationship (ER) Diagram

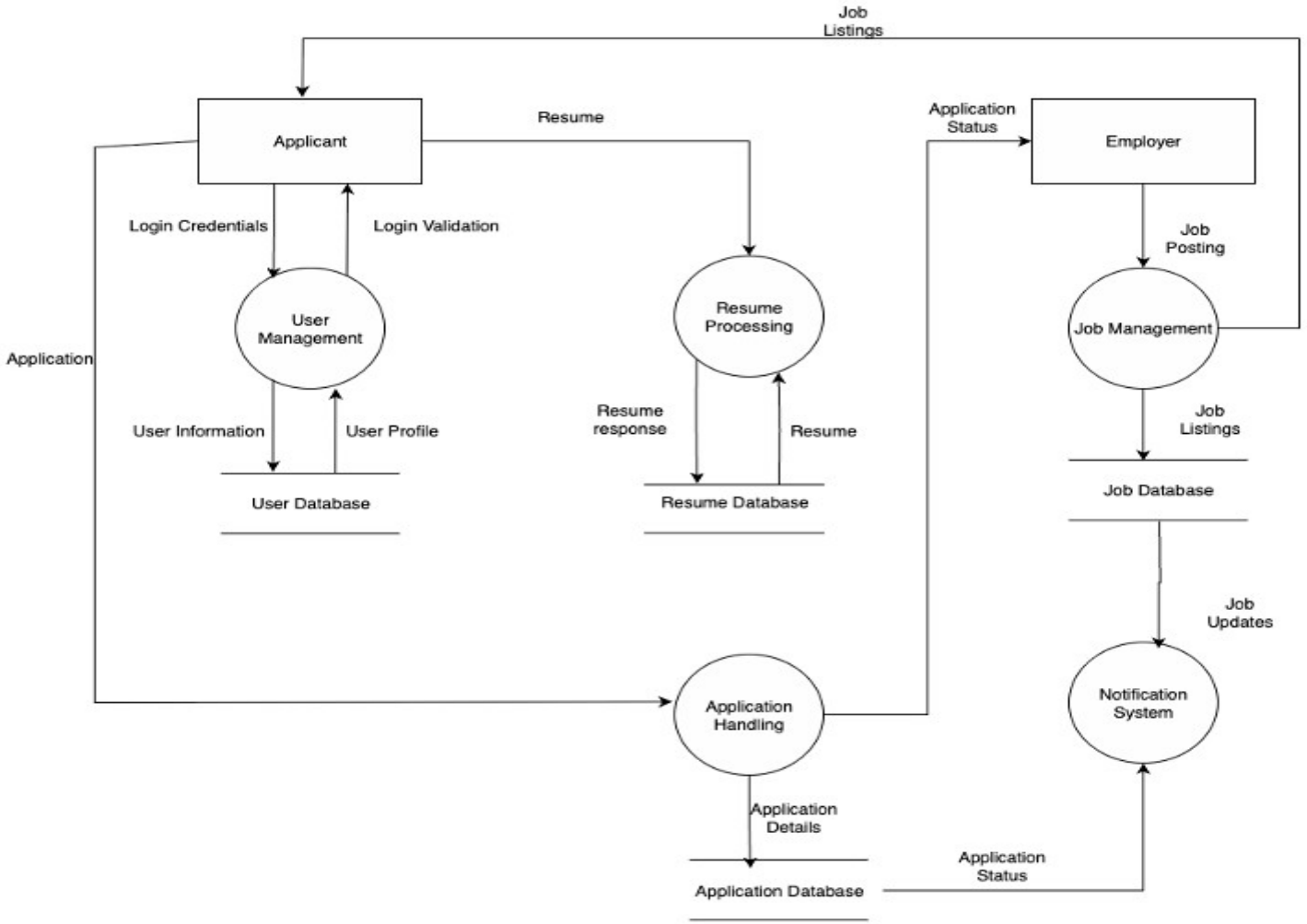


Data Flow Diagram (DFD)

- Context Level DFD (Level 0)

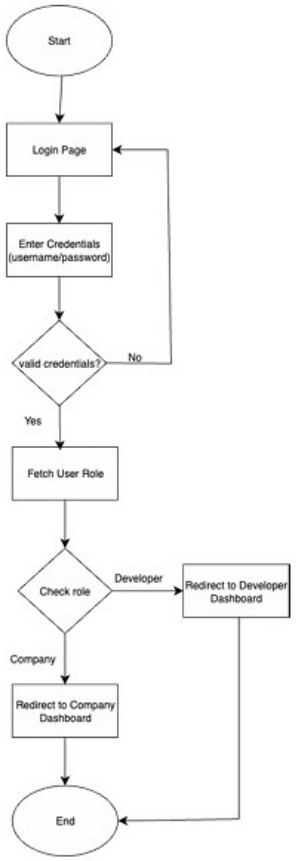


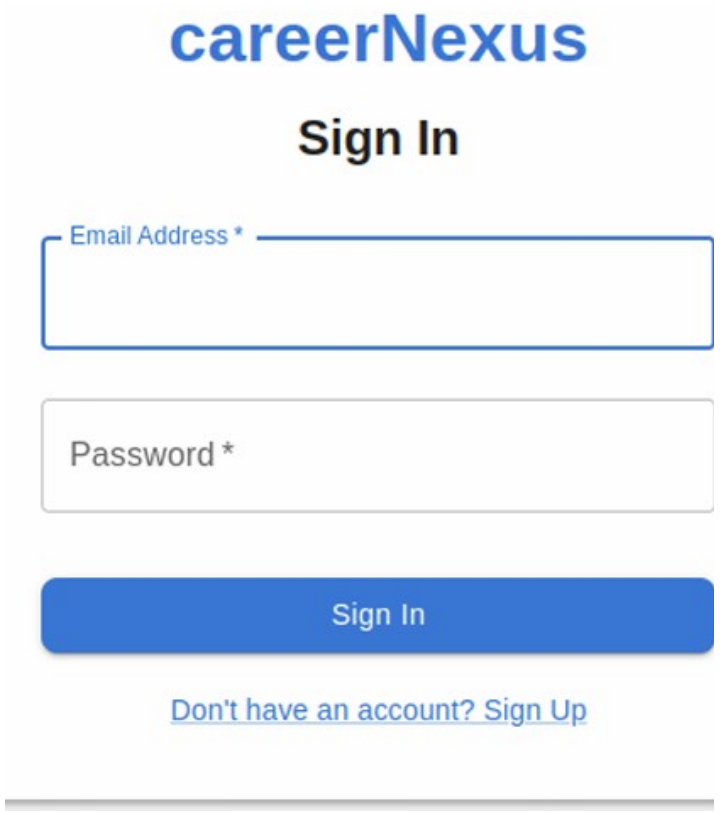
- Level 1 DFD


Job Listings
Applicant
Resume
Application Status
Employer
Login Credentials
Login Validation
Job Posting
User Management
Resume Processing
Job Management
Application
Resume response
Resume
Job Listings
User Information
User Profile
User Database
Resume Database
Job Database
Job Updates
Application Handling
Notification System
Application Details
Application Status
Application Database

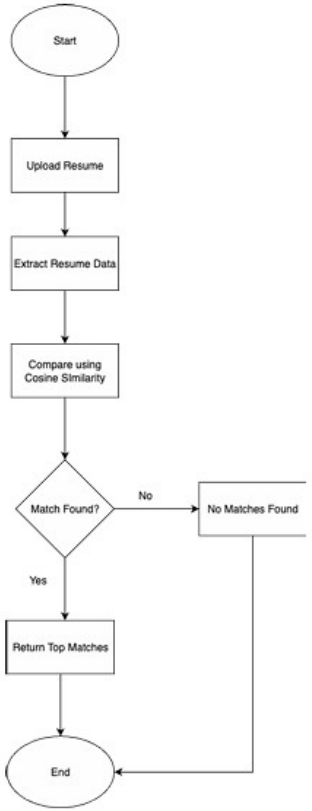
System Workflow

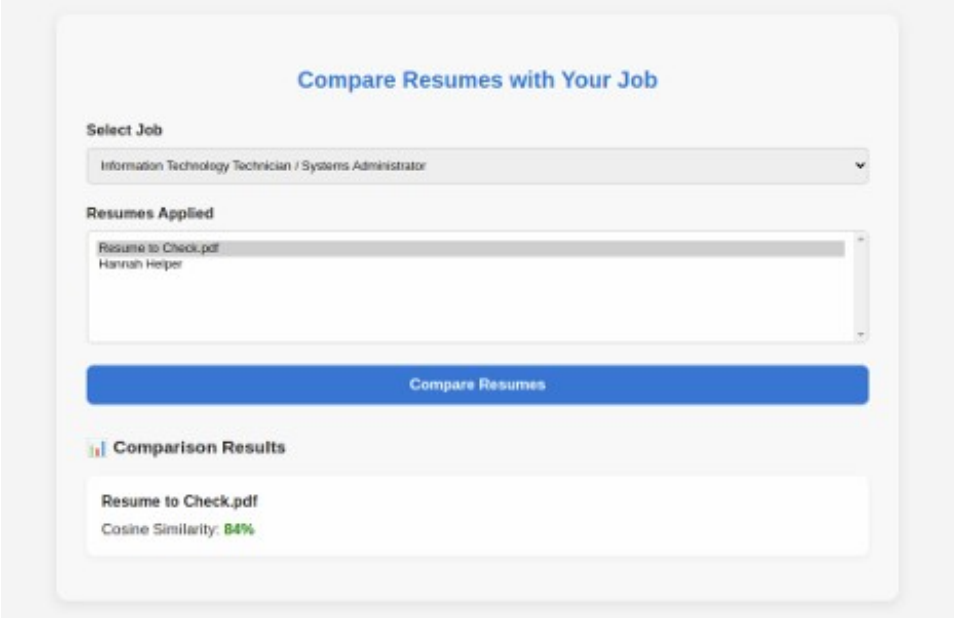
1) Sign in



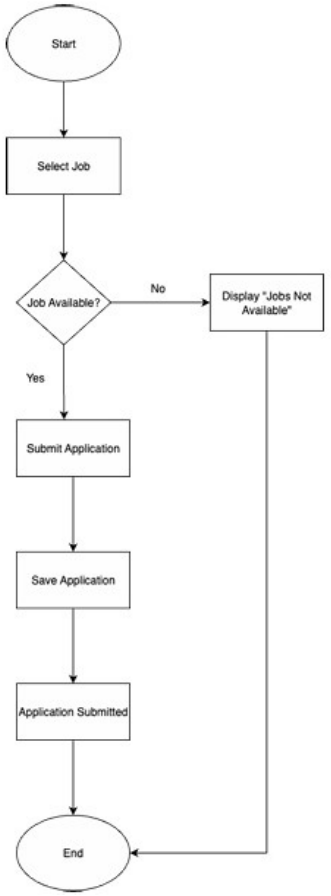


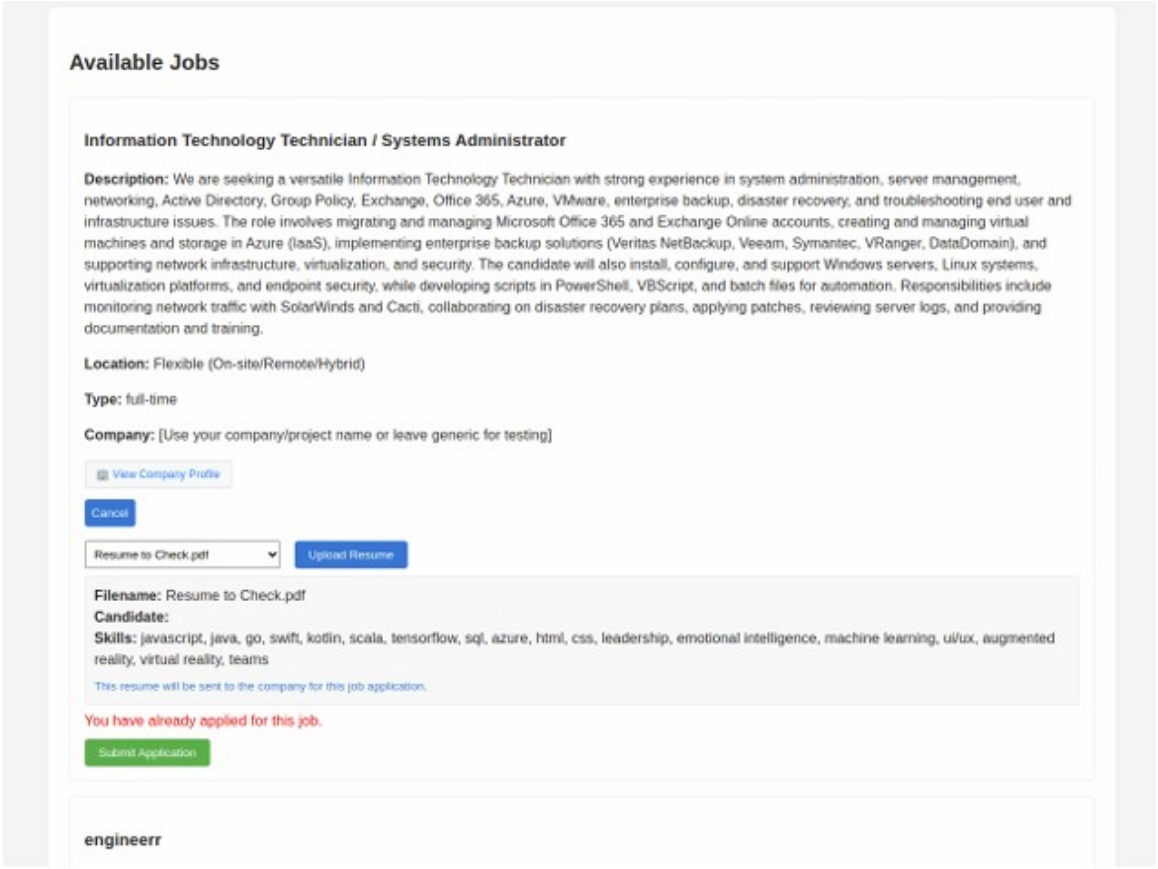
2) Resume Matching



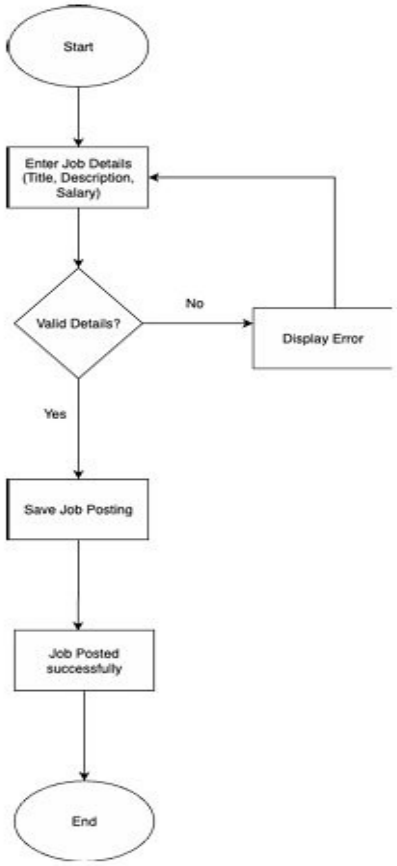


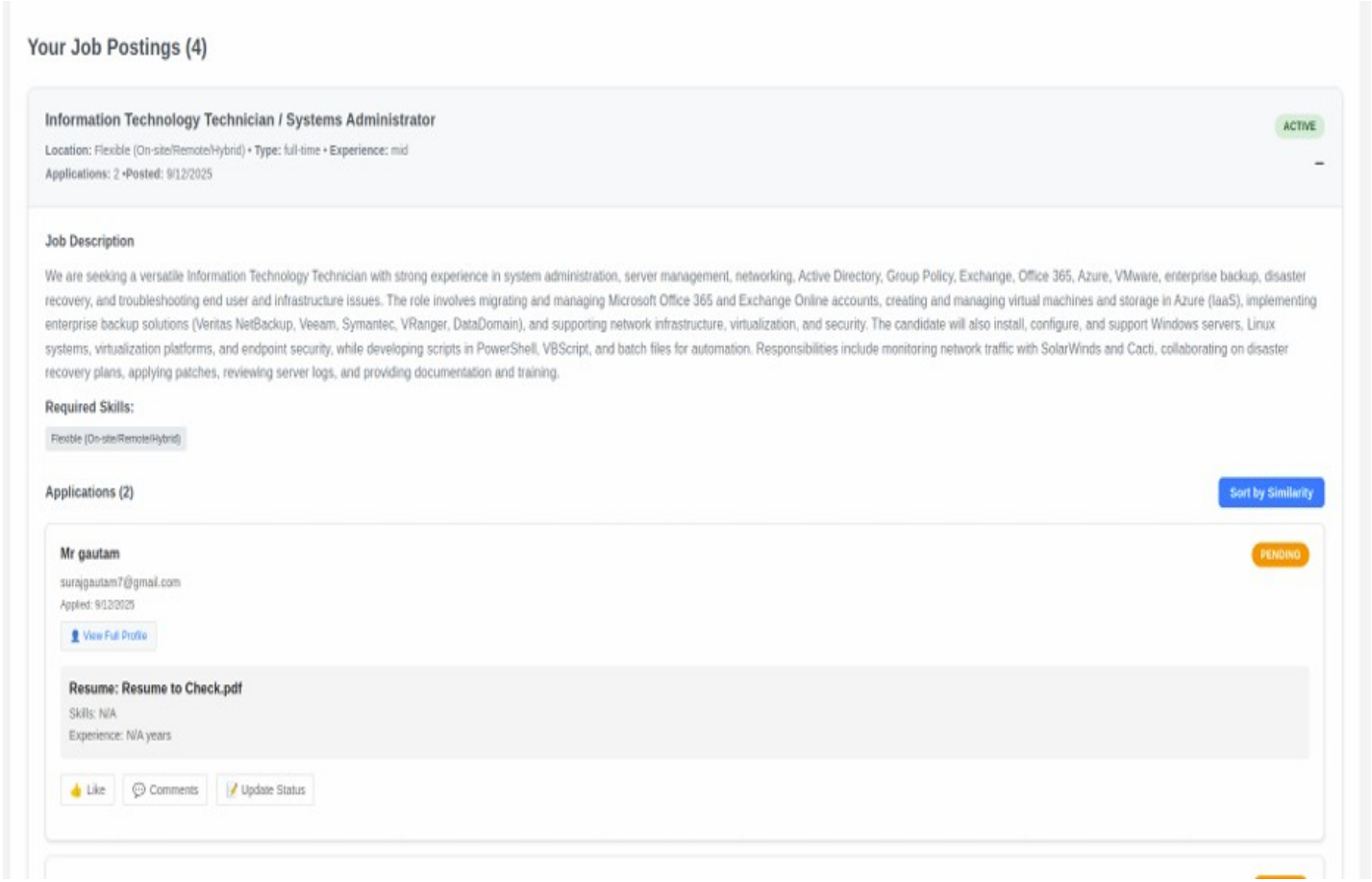
3) Job Application





4) Job Posting





Algorithm : Cosine Similarity

Definition :

Cosine similarity is a metric used to measure how similar two items (usually text documents or vectors) are, based on the angle between them in a multi- dimensional space

$$\text{Cosine Similarity} = \frac{A \cdot B}{\|A\|\|B\|}$$

where:

- $A \cdot B$ is the **dot product** of the vectors A and B.
- $\|A\|$ and $\|B\|$ are the **magnitudes** (lengths) of the vectors A and B respectively.

Implementation

This is the phase where our project design was actually coded, tested and deployed. We transformed the project blueprint into a functional software using necessary tools and technologies required to do so.

React JS :	Frontend Development
Express JS ex:	Backend Development
Node JS :	JavaScript Runtime
Mongo DB :	Database
VS Code IDE :	Coding and Development
Postman :	API Testing
Figma :	Designing wireframe and UI
GitHub :	Version Control

Testing

Testing is performed to investigate and validate the behavior of a fully integrated software product. Before deploying an application or website, it must be thoroughly tested.

Some of the types of testing that we did are described below:

Unit Testing : A unit test is a way of testing a unit - the smallest piece of code that can be logically isolated in a system. In most programming languages, that is a function, a subroutine, a method or property.

System Testing : System Testing plays a crucial role in ensuring that the entire system functions as intended and meets the specified requirements. System testing is done to check whether the software or product meets the specified requirements or not.

Result Analysis :

1) Login and Registration

Total Test Cases: 50

Passed: 48 (96 %)

Failed: 2

After Fix: 100% success

2) Resume Upload and AI Matching

Total Test Resume: 30

Correctly Matched: 26

Accuracy: 86.7%

3) Job Posting and Searching

Posting Time : Less than 2 seconds

Search Speed : Results appeared in 0.8 seconds on average

Relevance : 91% of search results matched user intent

Conclusion :

Career Nexus is a recruitment platform built with the MERN stack (MongoDB, Express JS, ReactJS, NodeJS) to provide efficient and precise matching between resumes and job openings using the cosine similarity algorithm. By overcoming the limitations of traditional keyword-based searches, it ensures a user-friendly experience for both job seekers and employers. The project offered valuable technical insights into modern web technologies, emphasized the importance of project management, iterative design, testing, scalability, and adaptability, and highlighted the need for effective team collaboration, knowledge sharing, and continuous improvement.

THANKYOU !!